Divide the signal entire

Group code: MON-04 Student names: Krunal Vaghela, Jahnvi Sharma, Sudhindra Lakse, Yashwardhan Khandegar, Vinay Hariharan EDL 2026

Project title: _____ Date: 19/01/2026

Use your notebook for discussions and rough work. Fill out this sheet after working individually and discussing within your team.

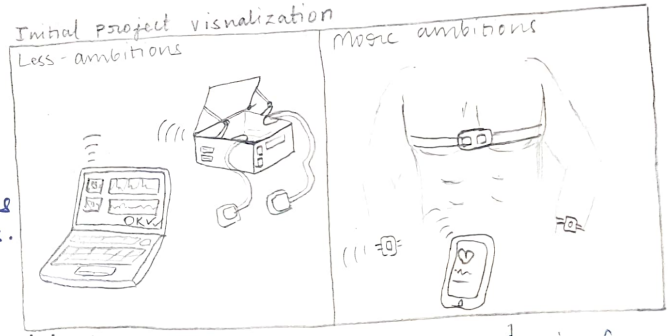
1. **In simple words, describe what you are going to build in your project, what its purpose is, and how it will function.** Be as detailed as possible, covering all the major aspects of your project.
   a. What is the main goal of your project?
   b. What problem does it solve, and how?
   c. Who will use your project, and in what context?

Draw a pencil sketch of what your project will look like at the end of the course, for final demo.

a. Ans: We want to make an accessible & portable medical device that would allow people to monitor health parameters (extracted from ECG & IPG signals) ~~the~~ for making quick & swift decisions.

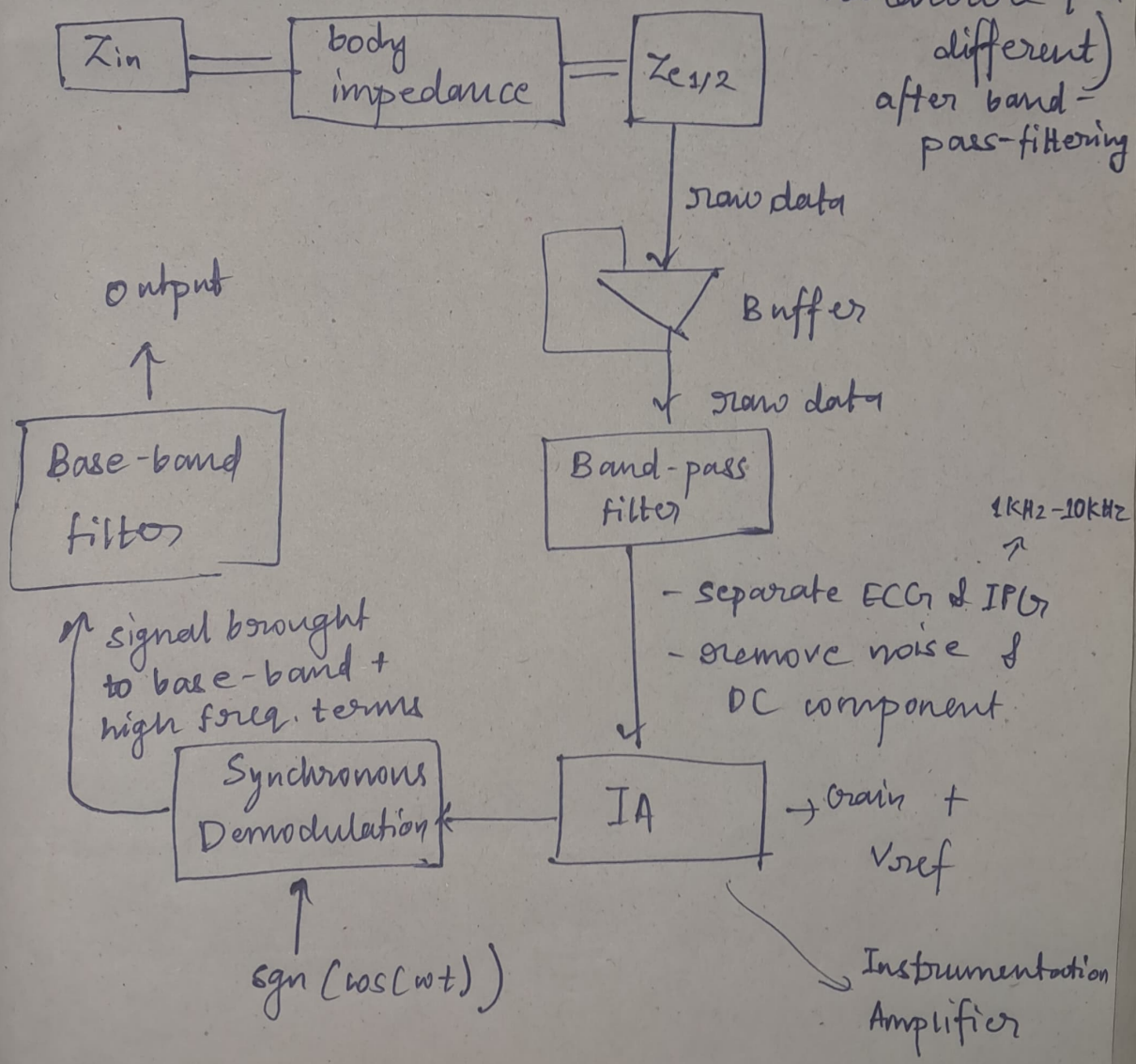
b. Ans: Currently we require separate equipments for obtaining ECG and bio-impedance measurements. And also they require multiple wet electrodes ton different locations such as the chest, legs which might be inconvenient to the users. This forms the main challenge that our project aims to tackle. Using just four electrodes (with two on each wrist) we aim to obtain both ECG and bio-impedance measurements. And wet electrodes are generally not re-usable unlike dry electrodes. We aim to use dry electrodes in our project. Our emphasis would also be on making a very user-friendly interface and calculate various health parameters from the extracted signals.

Initial project visualization



c. Ans: It would allow people both healthy as well as affected individuals monitor their bodily functions on a regular basis from home itself. Medical proffessionals visiting homes for check-up can carry it and use it for a quick and fairly accurate diagnosis. Data logging done by the device can be sent to a family doctor remotely or even fed to AI for analysis.

Initial Ideal of our project pipeline

Idea: Divide the signal extraction job between four electrodes in such a way that the two signals IPG & ECG interfere with each other to a minimum extent possible

ECG & IPG Pipeline: (General common to both but hardware path different) after band-pass-filtering